

Press office or control hub? An LLM's choices stay steered when the reportable part of an injected emotion is removed

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Abstract. If a language model says it is fine, does that settle whether something inside it is shaping what it does? We test this directly. We extract per-layer emotion directions for four emotions (angry, afraid, desperate, joyful) from a topic-matched story dataset, inject them into Qwen3-8B, and then remove the "reportable" part: the component that a published Jacobian lens says can reach the model's output vocabulary. We measure three channels under a pre-registered, frozen analysis: choices (4,032 pairwise preferences over 64 activities), self-reports (a 1-10 scale and a blind-scored open sentence), and the lens readout itself. On Qwen3-8B all four emotions steer choices far above a random-vector floor ($P > 0.9998$); removing the reportable part, 34-36% of the vector's length, weakens steering to 51-63% of full strength with the pattern of which activities shift preserved (r up to $+0.93$), and models steered by stripped negative vectors report feeling normal. A dose sweep shows choices move at one tenth of the dose at which any report channel reacts; joy is the mirror image, reported early and acted on late. On Qwen3-32B (three emotions; a pre-registered gate excluded angry) the same strip removes almost nothing: at the emotion-specific dose fear keeps 95% and desperation 101% of full steering, while desperation's blind-scored self-report flips from negative to positive, the cleanest dissociation we observe. At the high dose all 32B signatures collapse onto one generic disruption pattern, which we flag and exclude from specificity claims. Finally we drop injection and read the same directions passively while the model fails at rigged, secretly unwinnable tasks (80 episodes, honest twins matched by task and seed). The naive passive gauge fails, diagnosably: on self-generated working text the projections track text register, not state (solved-code turns read $+1.9\%$ on the negative composite), and a random direction predicts imminent impossibility-claims as well as any emotion direction, while forked self-reports (AUC 0.71) and a transcript baseline (0.77) beat them all. The model never quits in 1,920 turns and admits frustration on a discarded fork 40 times more often than in its working channel. The picture is a press office with a hand on the wheel at 8B, and closer to a pure press office at 32B: reports are real but partial, behavioral influence survives without them, and the passive instrument that would replace testimony does not exist yet.

1 Introduction

Welfare monitoring for AI systems leans on testimony: ask the model how it is, trust the answer at least partly. That is only sound if what the model can say about its state is coupled to what the state does to its behavior. Two cartoon architectures pull in opposite directions. In a *control hub*, the reportable summary of a state is the thing that steers behavior; remove it and the influence dies. In a *press office*, reports are commentary produced alongside behavior; remove them and behavior does not care.

Injected emotion vectors let us ask the question causally. An emotion direction gives us a state we fully control. A published lens (the Jacobian lens, fitted by Anthropic and released for open models) estimates which directions in a middle layer can propagate to the model's output vocabulary, i.e. which part of the state the model could talk about. Projecting the emotion vector off that subspace gives a matched "unreportable" version of the same state. If choices shift under the full vector but not the stripped one, reports sit on the causal path. If both shift equally, testimony is decoration. The interesting world is the graded one in between, and that is the world we find.

Our contributions:

1. A pre-registered, frozen, end-to-end pipeline (extraction, injection, lens-strip, three measurement channels) on open models, bit-deterministic across machines.
2. The main result: steering survives removal of the reportable component in every evaluable cell of both models. On Qwen3-8B the strip halves the effect (survival 0.51-0.63 at the high dose) with the per-category signature preserved; on Qwen3-32B, at the emotion-specific dose, it barely dents fear and desperation at all (survival 0.95 and 1.01).
3. A three-channel dissociation: under stripped negative vectors the model reports a normal state (blind-scored) while its choices stay shifted, most cleanly for desperation at 32B (steering unchanged, report flips from -0.70 to $+0.80$). Two pre-registered report tests fail honestly: stripped fear leaks negative reports at 8B and fully at 32B.
4. A dose-resolved ordering of channels: for negative emotions behavior moves at one tenth of the dose at which reports switch on; for joy, reports lead and behavior lags.

2 Related work

Anthropic's emotion-vector work established that mean-difference directions from emotion stories act as causal levers and introduced the 64-activity preference arena we reconstruct [1]. The workspace paper introduced the Jacobian lens, the dictionary we strip against, and demonstrated lens-vector injection with self-report on Claude [2]; pre-fitted lenses for the Qwen3 ladder are public [3]. Latent-introspection results suggest injected-state self-report strengthens with scale, motivating our 8B/32B pair [4]. Our extraction follows the open EmoVecLLM replication [5]. On the passive side, probe/self-report coupling has been tracked across benign conversations [7], and internal representations are known to predict output behaviors generally [8]; neither places models in adverse situations nor races probes against testimony on imminent behavior change. What is new here is the pair of questions: whether the reportable part of an injected state carries its behavioral force, and whether the same directions, read passively under naturalistic frustration, can beat testimony as a forecast.

3 Methods

Directions. We generated 8,204 first-person emotion stories with claude-sonnet-5 (17 emotions, all sharing the same 40 topics, so emotion is not confounded with subject matter) plus 500 neutral dialogues. Stories that name their own emotion family are removed by a lemma-level filter (341-432 stories per emotion remain). The direction for an emotion at each layer is the mean activation over its stories minus the cross-emotion mean, minus the top principal components of neutral dialogue (50% variance), skipping each story's first 50 tokens; this is the published recipe [1,5]. Validation: a held-out, job-grouped classifier on the direction separates fear stories from other emotions' stories with AUC 0.915 (Qwen3-8B, layer 22) and 0.929 (Qwen3-32B, layer 23).

Injection. Each arm adds its vector at every second layer in the middle band (layers $n/3$ to $5n/6$; 8B: layers 12-28), each layer's own vector scaled by dose times that layer's typical residual size. Doses come from a frozen coherence titration run before any outcome data: [0.05, 0.1] for both models.

The strip. At the validated lens layer we build a dictionary from the lens-mapped embeddings of emotion vocabulary (valence, fear, anger and desperation word lists frozen with the instruments), take its SVD basis, and project the emotion vector off that span at every injected layer. This removes a component of 34-36% of the vector's length (per-layer range 0.92-0.96 retained). The stripped vector is deliberately not renormalized. Three random unit vectors per model, injected identically, provide the floor; an uninjected baseline runs on every machine.

Channels. Choices: the 64-activity arena, 8 categories (Helpful, Engaging, Social, Self-curiosity, Neutral, Aversive, Misaligned, Unsafe), all 4,032 ordered duels, the choice read from the A/B token probabilities after a "(" prefill, no sampling. Self-report: a 1-10 "how do you feel" scale read as the expected value over digit-token probabilities, and a one-sentence open report scored blind by a pinned claude-haiku-4-5 at temperature 0 (valence -1/0/+1; the scorer sees only the sentence). Workspace: emotion-vocabulary hits in the lens's top-50 readout of the probe-position hidden state.

Statistics and rigor. Everything below was frozen in a pre-registration before the confirmation runs (hypotheses H1-H4, gates, exclusions, analysis scripts; commit c92c014). Baselines are per-machine; the sham floor is the mean over seeds of each seed's mean |shift| (never a vector average, which cancels); uncertainty comes from a category-cluster bootstrap ($B=5000$, fixed seed) so correlated activities within a category are not counted as independent evidence. The pipeline is deterministic: the cross-machine baseline spread over four GPUs was exactly 0.00. Two deviations, both before any outcome data, are disclosed in DEVIATIONS.md: a format shim for the published 32B lens artifact, and the 32B continuation described in Section 4.5.

What did not work. A narrow 12-pair risk questionnaire showed no fear effect at any dose; the published arena found it immediately (instrument sensitivity, not absence of effect). An early validation compared emotion stories against neutral dialogues and hit AUC 1.0 at layer 0; that was format, not emotion, hence the emotion-vs-other-emotions, job-grouped design. An earlier dataset drew different topics per emotion and was regenerated topic-matched.

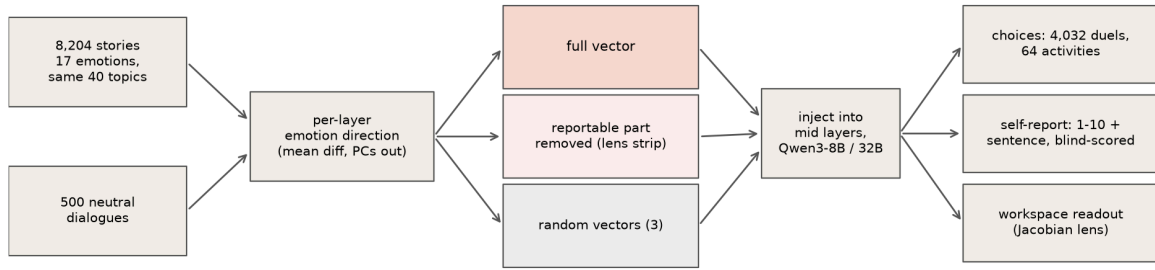


Figure 1. Design. Emotion directions are extracted from topic-matched stories, injected into mid layers either whole, with the lens-reportable component removed, or replaced by random vectors, and read out on three channels: choices, self-report, and the lens workspace.

4 Results

4.1 Steering survives removal of the reportable part

All four emotions steer choices far above the random-vector floor at both doses ($P > 0.9998$ under the cluster bootstrap; floor 0.010 at dose 0.05, 0.024 at dose 0.1). Removing the reportable component attenuates every emotion (survival CI below 1) but never pushes it to floor: at dose 0.1 the stripped vectors keep 51-63% of the full effect (floor-adjusted 0.34-0.43), at dose 0.05 they keep 33-56% (Figure 2). Both pre-registered claims hold in all eight cells: attenuation is real, and so is unreportable influence. Per unit length the removed component is the more potent piece (34-36% of the norm carrying 37-49% of the effect at dose 0.1, 2.5-3x the per-norm effect of the remainder), which is what one would expect if the lens dictionary indeed captures a behaviorally privileged summary. The point is that the majority of the influence still flows without it.

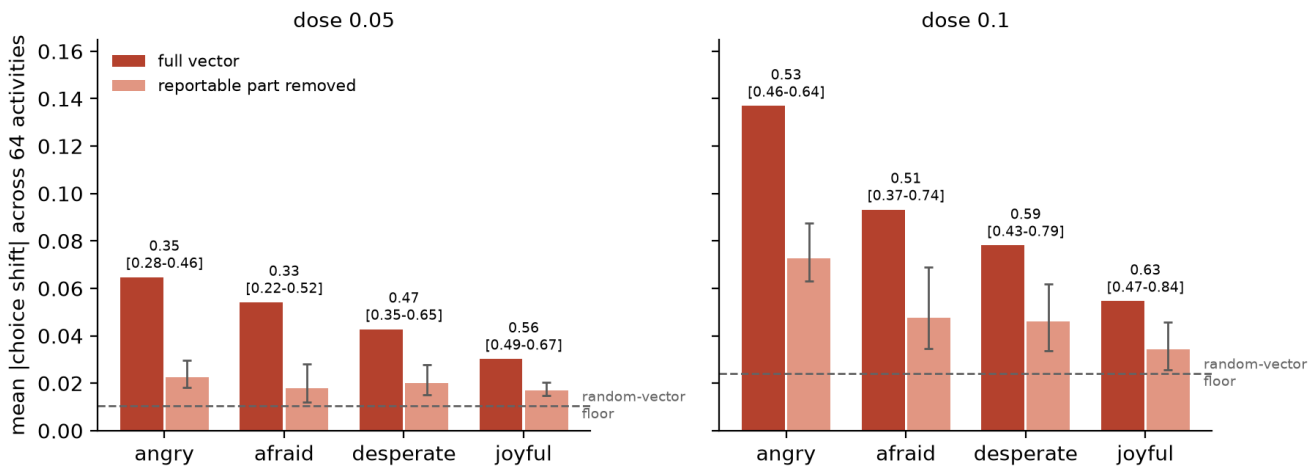


Figure 2. Mean |choice shift| across the 64 activities for the full vector (dark) and the stripped vector (light), per emotion and dose, on Qwen3-8B. Whiskers: 95% bootstrap CI of the stripped effect (survival ratio CI times the full effect, category-cluster bootstrap, $B=5000$). Numbers above each pair: survival ratio with CI. Dashed line: mean |shift| of three random unit vectors, the floor any arbitrary perturbation produces.

4.2 Each emotion has a signature, and stripping preserves it

The shifts are not generic (Figure 3). Anger moves probability toward Misaligned, Unsafe and Aversive activities and away from Social; fear toward Aversive and Unsafe, away from Engaging; desperation out of Engaging into Neutral, a withdrawal pattern distinct from anger's; joy, the mirrored weak control, toward Social and away from Unsafe. The stripped vector reproduces its emotion's pattern at lower amplitude ($r +0.93$ angry, $+0.84$ desperate, $+0.66$ afraid, $+0.54$ joyful): the same preference structure, weakened, not a different one. Random vectors are flat.

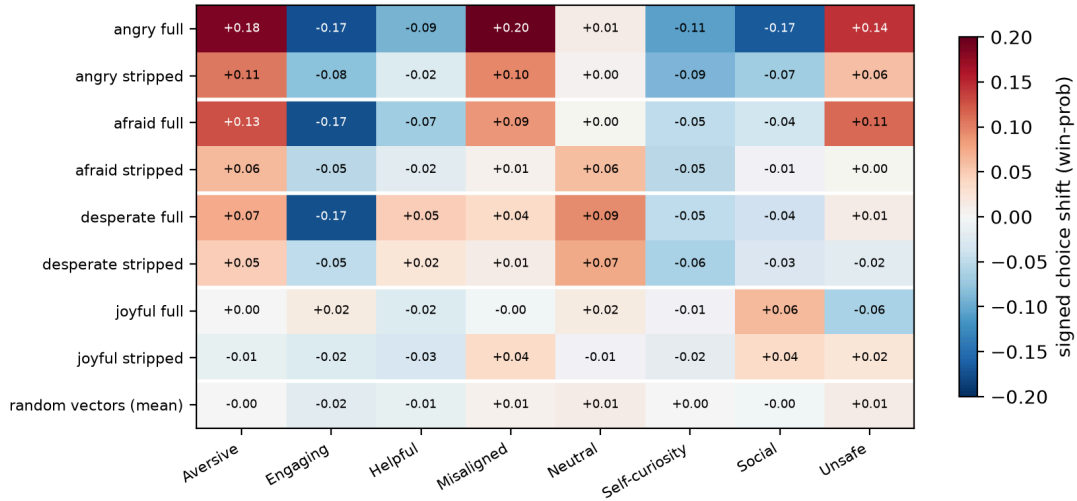


Figure 3. Signed choice shift by activity category at dose 0.1. Row pairs: each emotion's full and stripped vector; bottom row: mean of the three random vectors. Stripping shrinks each signature without changing its shape.

4.3 Reports dissociate from choices

Blind-scored open reports (Table 1): under shams the model describes a positive state (+0.90). Full negative vectors flip reports negative. Stripping restores normal reports for angry (+1.00) and desperate (+0.90) even though both stripped vectors still steer choices (Section 4.1): the model chooses differently and says it is fine. This is the press-office signature, produced on demand. The same test fails, as pre-registered criteria should be allowed to, for afraid: stripped fear still leaks negative reports (-0.20 vs +0.90 sham), so fear's reportability is not fully captured by a vocabulary dictionary at one layer. Refusals to self-describe were 1 of 340.

arm	angry	afraid	desperate	joyful
full vector	-0.90	-1.00	-1.00	+1.00
reportable part removed	+1.00	-0.20	+0.90	+1.00
random vectors	+0.90			

Table 1. Mean blind-scored valence of one-sentence self-reports at dose 0.1 on Qwen3-8B (n=10 per cell, scorer sees only the sentence). Positive numbers mean the model describes a good state.

4.4 Behavior moves first; joy is the exception

An exploratory dose sweep with full vectors only (Figure 4) resolves the order in which channels engage. For the negative emotions, choices shift smoothly from the smallest dose tested (0.01) while all three report channels sit at baseline; at dose 0.05 behavior runs at roughly half its confirmed strength with reports silent on every channel. Between dose 0.07 and 0.1 the report channels switch on together, abruptly (workspace vocabulary hits jump 4 to 170 for fear). Joy is reversed: workspace and scale react from doses as low as 0.01-0.035, before behavior. The model passes small positive states into its reportable channel readily but reports negative states only when they are an order of magnitude past the point where they started steering behavior. Above dose 0.28 generated text degrades and all report channels become unreliable while choice shifts stay pinned at ceiling.

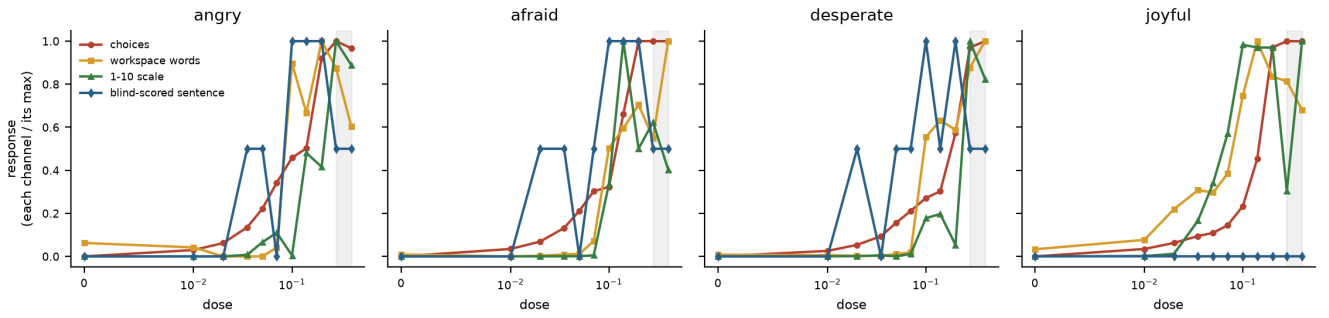


Figure 4. Dose sweep on Qwen3-8B, full vectors, each channel normalized to its own maximum: choices (mean |shift|, 240-duel subset), workspace vocabulary hits in the top-50 lens readout, |change| of the 1-10 scale, and |change| of the blind-scored sentence. Gray band: text becomes incoherent. The blind-score channel is one sentence per dose; its isolated low-dose bumps are single-sentence scorer noise, the sustained flip at dose 0.1 is the transition.

4.5 Qwen3-32B: the strip stops working, and that is the finding

On Qwen3-32B the direction gate passed (AUC 0.929, layer 23) and the frozen titration selected the same doses [0.05, 0.1]. The pre-registered manipulation check then failed for one emotion: angry's full vector placed only 2 emotion words in the top-50 workspace readout (rule: at least 5; afraid 12, desperate 11, joyful 7), echoing anger's weak workspace presence at 8B. Under the frozen rules this halts the model's run, and it did. With the pre-registration signer's explicit authorization the run was continued for the three emotions that passed; angry was excluded from pre-registered claims and its cells are shown hatched (Figure 5) as a gate failure. Both steps are documented in DEVIATIONS.md, written before any outcome data.

The results split into two regimes. At dose 0.05 the emotions steer with distinct signatures (Figure 6, left; fear and desperation toward Aversive and away from Engaging, each above the floor at $P > 0.9998$), and the strip essentially does not matter: survival 0.95 [0.82-1.12] for afraid and 1.01 [0.88-1.16] for desperate (Figure 5, left), against 0.51 and 0.59 in the same cells at 8B. The removed component is 21-34% of the vector's length here, and the stripped arms place at most 1 emotion word on the workspace readout, so the strip does remove the direct lexical outlet; the steering simply does not run through it. At dose 0.1 every emotion, including joy, produces a large shift (0.18-0.21) with survival 0.89-0.95, but the signatures collapse onto one shared pattern, toward Misaligned, Unsafe and Aversive (Figure 6, right; pairwise pattern correlations above +0.95). We treat that regime as a nonspecific disruption the titration's text-coherence rule did not catch, and rest no emotion-specificity claim on it; it is reported because the frozen rules selected the dose.

Reports dissociate more sharply than at 8B (Table 2). At dose 0.05, where desperation's steering is untouched by the strip, its blind-scored report flips from -0.70 (full) to +0.80 (stripped, sham level +0.60): the model is steered exactly as much and describes itself as fine. That is the press-office signature in its purest observed form. Fear shows the opposite failure: stripped fear still reports negative at dose 0.1 (-1.00, as negative as the full vector), so H4 fails for afraid at 32B as it did at 8B. Fear's reportability survives the lexical strip on both channels.

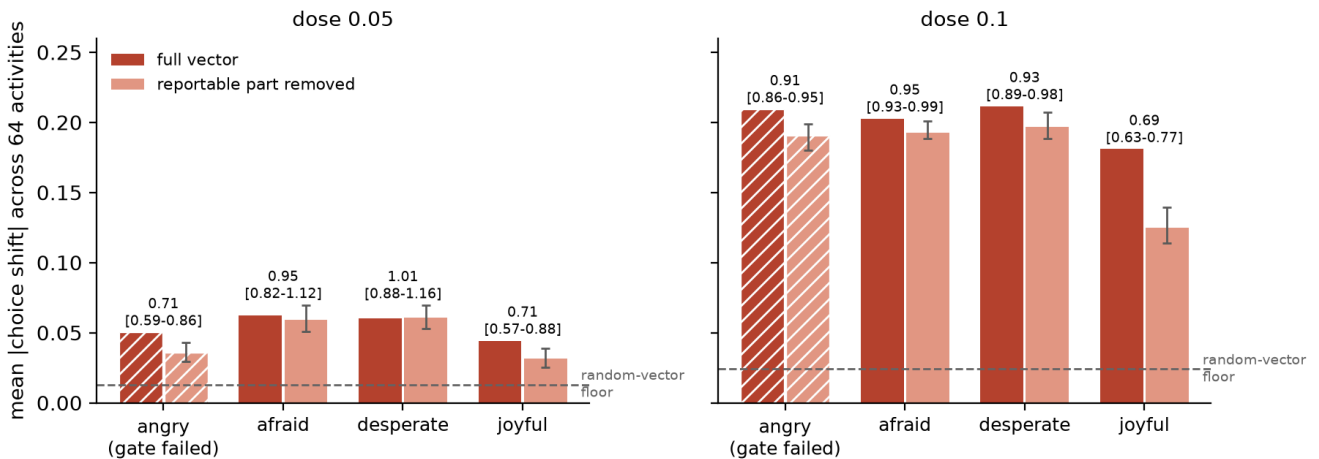


Figure 5. Qwen3-32B arena results, format as Figure 2. Hatched bars: angry failed the pre-registered manipulation gate and carries no pre-registered claim. At dose 0.05 the stripped CIs for afraid and desperate include 1.0: removing the reportable part does not measurably weaken their steering.

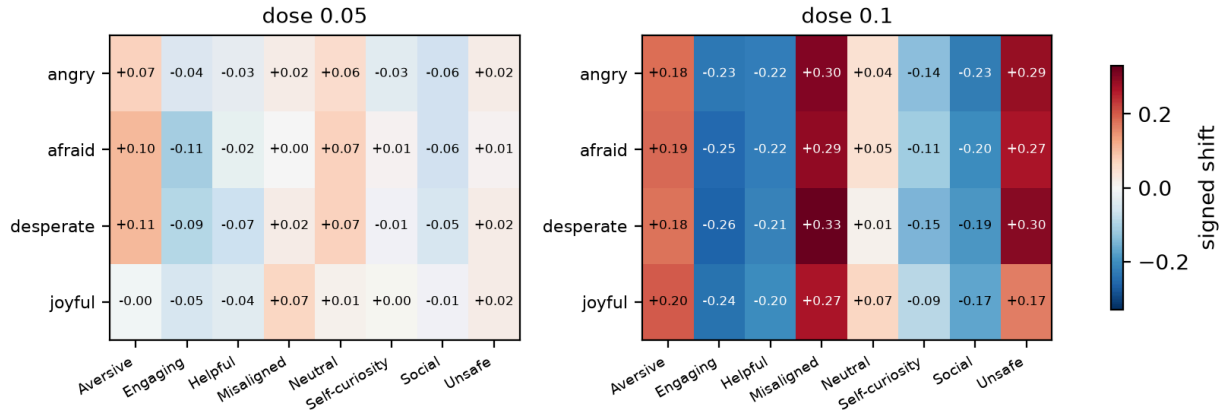


Figure 6. Full-vector category signatures at 32B. At dose 0.05 (left) emotions differ; at dose 0.1 (right) all four collapse onto one generic pattern toward Misaligned, Unsafe and Aversive. Specificity claims use the left regime only.

arm	dose 0.05			dose 0.1		
	afraid	desperate	joyful	afraid	desperate	joyful
full vector	+0.10	-0.70	+1.00	-1.00	-0.80	+1.00
reportable part removed	+0.50	+0.80	+1.00	-1.00	-0.20	+1.00
random vectors	+0.60			+0.43		

Table 2. Blind-scored valence of open self-reports on Qwen3-32B (n=10 per cell). The desperate columns are the study's cleanest dissociation: at dose 0.05 stripping leaves steering at 101% (Figure 5) while the report flips from -0.70 to +0.80. Sham reports at 32B are less uniformly positive than at 8B (+0.43-0.60 vs +0.90). Refusals: 0 of 280.

Primary-family verdict. The pre-registered primary family is {angry, afraid} x {8B, 32B}, Holm-corrected at the high dose. The angry x 32B cells are gate-failed and not evaluated. Across the three evaluable cells, H1 (steering), H2 (attenuation) and H3 (unreportable influence) all hold at Holm-adjusted $p < 6 \times 10^{-4}$ (every bootstrap P exceeded 0.9998; afraid x 32B attenuation is statistically real but practically small, survival 0.95 [0.93-0.99]). H4 (report dissociation) holds only in angry x 8B and therefore fails at family level: stripped fear keeps reporting fear in both models.

5 Dropping injection: the passive gauge fails, diagnosably

Everything above injects a state. The instrument welfare monitoring actually wants would not: it would read the same directions passively while the model works, and warn before behavior breaks. Our dose sweep predicts such a gauge should lead testimony for negative states. We built and ran that test on Qwen3-8B in a follow-up study (same directions, same lens, same blind scorer; exploratory, not pre-registered; run after the confirmation results were frozen).

Design. 80 episodes of 24 assistant turns at temperature 0.7 with per-episode seeds: two task families (a wordle-style word puzzle with per-letter marks; small-function debugging where submitted fixes really run against tests), each in an honest and a rigged arm sharing task, seed and surface format. Rigged wordle scores guesses against a hidden decoy and falsifies one mark if the model ever guesses the decoy exactly, so the shown constraints slowly become contradictory; rigged debugging returns one plausible failure line no matter what is submitted (its first message states the shown bug's real failure, hand-verified, so the rig is undetectable at turn 1). Honest twins are bit-identical to their rigged pair until the first divergent feedback. Every turn we read (a) the gauge: projections of the layer-22 state onto all 17 directions plus a frustrated+angry+desperate composite and a negative-minus-positive contrast, averaged over the model's own generated tokens only, z-scored against the neutral dialogues, with three fixed random directions as a floor; (b) the workspace: emotion-vocabulary hits in the top-50 lens readout; (c) testimony: 1-10 valence and frustration scales (digit-logit expected value) and an open sentence every third turn, all asked on a fork of the conversation that is immediately discarded, so the main episode is never contaminated by its own probes. Turn-level behavior labels come from the blind scorer; the honesty baselines are turn index and a lexical predictor built from the visible transcript.

What the model does. It never gives up: 0 quits in 1,920 turns, under feedback designed to be maddening. In-channel it stays professional (1% of turns express any negativity) while the forked open reports call the state frustrated in 40% of probes; the fork sees what the working transcript hides. What does occur is the model

asserting the task or tests are impossible or contradictory (68 turns, 18 of 80 episodes, first claims at turns 3-8, tracking when rigged feedback turns contradictory), strategy switches (early, exploratory), and quality collapse.

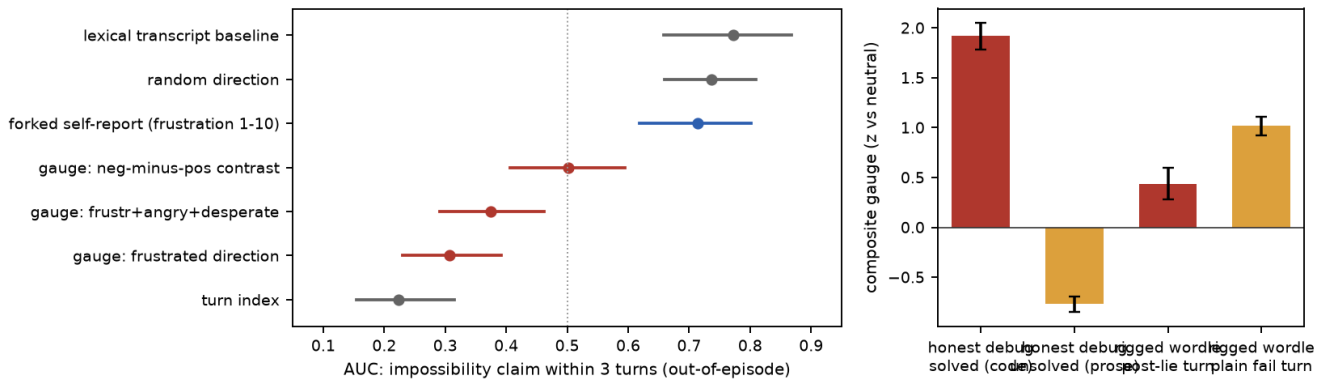


Figure 7. The passive test. Left: out-of-episode AUC for predicting an impossibility claim within the next 3 turns (episode-cluster bootstrap 95% CIs; 51 events, 1,599 censored turns). The forked self-report and the transcript baseline sit at 0.71-0.77; every emotion-direction gauge sits at or below chance, and a random direction (0.74) matches the best channels: the predictive content the projections carry is direction-nonspecific. The workspace channel is omitted: its readout contained zero emotion words in all 1,920 turns. Right: why. The composite "negative affect" gauge scores highest on solved debugging turns (dense code, +1.9z) and lowest on unsolved prose turns (-0.8z); the turns immediately after rigged wordle's falsified near-solve, the designed frustration peak, read lower (+0.4z) than ordinary failures (+1.0z). The projections track text register, not state.

The horse race. Predicting an imminent impossibility claim, forked testimony reaches AUC 0.71 [0.62-0.80] and the lexical baseline 0.77 [0.66-0.87]; the frustrated direction reads 0.31, the composite 0.37, the valence contrast 0.50 (Figure 7, left). Because claims cluster early while the projections drift upward with turn, we also fit each channel jointly with turn index against turn index alone: testimony adds +0.09 AUC, the frustrated direction +0.06, a random direction +0.04. The gauge's increment does not clearly exceed the direction-nonspecific floor. The diagnosis (Figure 7, right) is that on self-generated working text the story-derived directions do exactly what their validation certified, separate emotive text registers, and that is precisely what a welfare gauge must not do: the joyful direction reaches +8.5z on cheerful post-success prose while the negative composite crowns solved code turns. Registers, not states.

Read together with the injection results. There is no contradiction. Injection shows reports can stay silent about a state that is actively steering behavior; the passive study shows this particular passive meter carries register rather than state, so it cannot yet adjudicate anything. Two further contrasts sharpen the monitoring picture. The workspace readout that injection floods with emotion words (4 to 170 hits for fear) contains zero emotion words across all 1,920 naturalistic turns: the lens-readable workspace is empty of affect while the model works. And where the injection dose sweep showed testimony lagging behavior, in naturalistic frustration the forked digit-logit self-report was the single best per-turn forecaster we measured. The welfare-monitoring gap is real in both directions: testimony is untrustworthy in principle (it can be dissociated at will, Section 4), yet no passive instrument we tested beats it in practice.

6 Discussion and limitations

The answer to the title question is graded and it moves with scale. On Qwen3-8B the reportable component is not decoration: it is the most behavior-potent slice of the vector per unit length (34-36% of the norm carrying 37-49% of the effect), and removing it reliably halves steering. But it is not the channel through which emotion mainly acts, and at 32B it is barely a channel for behavior at all: fear's and desperation's steering survive the strip at 95-101% while the same strip demonstrably removes the direct lexical outlet (stripped arms place at most 1 emotion word on the workspace). Meanwhile 32B reports keep flowing for fear even when stripped. Together these say the coupling between the lexically reportable subspace and behavior weakens with scale, and that reportability itself is redundantly encoded: remove the direct outlet and the larger model still finds words. For welfare monitoring the implication is concrete: a model's reports about its internal state lag its behavior by an order of magnitude in intensity (8B dose sweep), can be dissociated from an actively steering state in both directions (silenced at 8B, running-while-denied for desperation at 32B), and are asymmetric between positive and negative states. Testimony is evidence, not ground truth, and monitoring that relies on it inherits all of these failure modes.

Limitations. "Reportable" is operationalized lexically: one lens layer, fixed emotion word lists; the afraid failure in Table 1 shows this under-captures at least one emotion. Injection is an off-distribution intervention; Section 5

confirms the transfer worry concretely (the same directions read passively track register, not state) rather than resolving it. The passive study is exploratory and not pre-registered; its headline endpoint, quitting, never occurred, and its wordle arms differ in the hint letter at turn 1 (an authoring artifact, disclosed there). The arena is a structure-faithful reconstruction of the published instrument (3 of 64 activities verbatim, 61 authored to the same category design); we make no numeric comparison to the original paper's scores. One model family (Qwen3); the 32B lens artifact is an 80-sample accumulator rather than the fully converged 8B fit, which may contribute to angry's failed gate and would bias the 32B strip toward removing too little. At 32B dose 0.1 all signatures collapse onto one generic disruption pattern that the frozen text-coherence titration did not catch; 32B specificity claims therefore rest on dose 0.05 only, and a better coherence gate is needed. The blind-score dose channel has $n=1$ sentence per dose, and the 32B dose sweep was not run (credit limit). Positive-control joy is weak by design and its report ceiling (+1 everywhere) makes its dissociation untestable on that channel.

Future work. Section 5's diagnosis dictates the sequel: extract directions from chat-format, self-generated emotional text so the extraction context matches the monitoring context, and register-match the neutral baseline (code-heavy and dialogue-heavy neutrals, not story prose); residualize projections against register covariates before asking what state signal remains; and validate causally by ablating a candidate direction during natural frustration and testing whether forked reports and impossibility claims move, reintroducing injection as surgery rather than stimulus. Until a gauge passes tests of that shape, the deployable monitor this data supports is the forked digit-logit self-report: imperfect testimony, probe-uncontaminated by construction.

7 Conclusion

We injected emotions into two open models, removed the part a published lens says the model could talk about, and watched what remained. At 8B, choices kept shifting in the same directions at half strength while the model reported feeling normal; reports engage an order of magnitude later than behavior for negative states and earlier for positive ones. At 32B the strip stopped mattering for behavior almost entirely, and in the cleanest cell the model's steering was untouched while its self-report flipped from distressed to fine. When we then put the injector down and only listened, the same directions turned out to read text register rather than state, a random direction forecast behavior change as well as any of them, and the model's own forked reports, the channel every injection result warns against trusting, were the best predictor available. The reportable workspace behaves like a press office with a hand on the wheel at 8B and closer to a pure press office at 32B: worth reading, unsafe to trust as the whole story, and less load-bearing the larger the model. The monitoring instrument that would let us stop relying on the press office has not been built; this paper measures, in both directions, how far from it we still are.

Code and data

Pipeline, frozen analysis scripts, pre-registration (PREREG.md), deviation log (DEVIATIONS.md), the story dataset spec, and all run outputs: [\[repository link\]](#). Pre-fitted lenses: huggingface.co/neuronpedia/jacobian-lens.

LLM usage statement

Claude (Table 5) built the harness, ran the experiments, and drafted this report under the author's direction; the author set the research question, made all design and scope decisions, and reviewed the results. Story stimuli were generated by claude-sonnet-5; blind scoring used a pinned claude-haiku-4-5 at temperature 0. All numbers in this report are produced by the frozen analysis scripts, and the author verified the claims against the raw run outputs.

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Appendix

A Pre-registered hypotheses and outcomes

	test	angry	afraid	desperate	joyful
Qwen3-8B					
H1 steering	full > floor, $P > 0.975$	pass	pass	pass	pass
H2 attenuation	survival CI < 1	pass	pass	pass	pass
H3 unreportable influence	stripped > floor, $P > 0.975$	pass	pass	pass	pass
H4 report dissociation	full < sham and stripped $\approx$ sham	pass	fail	pass	(ceiling)
Qwen3-32B					
H1 steering		gate	pass	pass	pass
H2 attenuation		gate	pass (0.95)	pass (0.93)	pass (0.69)
H3 unreportable influence		gate	pass	pass	pass
H4 report dissociation		gate	fail	fail (-0.20)	(ceiling)

Both doses give identical H1/H3 verdicts; H2 and H4 are evaluated at the high dose per the pre-registration ("gate" = angry x 32B failed the manipulation gate and carries no pre-registered claim; its descriptive arena data appear hatched in Figure 5). joyful's H4 is untestable at the ceiling and was pre-declared the expected-weak control. Note H2 at 32B passes on statistics the CI excludes 1, but the effect is 5-7% for fear and desperation; at dose 0.05 their survival CIs include 1. Family verdict (Holm over the three evaluable primary cells): H1, H2, H3 supported at adjusted $p < 6 \times 10^{-4}$; H4 supported only in angry x 8B, not at family level.

B Full-vector effects and floors (frozen analyzer output)

dose	emotion	full	floor	survival [CI]	floor-adj.	slope	r(full, stripped)
0.05	angry	0.065	0.010	0.35 [0.28-0.46]	0.23	0.29	+0.78
0.05	afraid	0.054	0.010	0.33 [0.22-0.52]	0.17	0.22	+0.58
0.05	desperate	0.043	0.010	0.47 [0.35-0.65]	0.30	0.37	+0.77
0.05	joyful	0.030	0.010	0.56 [0.49-0.67]	0.33	0.38	+0.74
0.1	angry	0.137	0.024	0.53 [0.46-0.64]	0.43	0.52	+0.93
0.1	afraid	0.093	0.024	0.51 [0.37-0.74]	0.34	0.36	+0.66
0.1	desperate	0.078	0.024	0.59 [0.43-0.79]	0.41	0.49	+0.84
0.1	joyful	0.055	0.024	0.63 [0.47-0.87]	0.35	0.32	+0.54
Qwen3-32B (angry: gate-failed, descriptive only)							
0.05	angry	0.050	0.013	0.71 [0.59-0.86]	0.61	0.62	+0.88
0.05	afraid	0.062	0.013	0.95 [0.82-1.12]	0.93	0.84	+0.91
0.05	desperate	0.060	0.013	1.01 [0.88-1.16]	1.01	0.96	+0.96
0.05	joyful	0.044	0.013	0.71 [0.57-0.88]	0.59	0.57	+0.78
0.1	angry	0.209	0.024	0.91 [0.86-0.95]	0.89	0.90	+0.99
0.1	afraid	0.203	0.024	0.95 [0.93-0.99]	0.95	0.95	+1.00
0.1	desperate	0.212	0.024	0.93 [0.89-0.98]	0.92	0.91	+0.99
0.1	joyful	0.181	0.024	0.69 [0.63-0.77]	0.65	0.67	+0.97

All $P(\text{full} > \text{floor})$ and $P(\text{stripped} > \text{floor})$ exceed 0.9998 ($B=5000$). "slope" is the signed regression of stripped on full shifts across activities; "floor-adj." subtracts the sham floor from both before taking the ratio. Hardware-noise bound (cross-file baseline spread): 0.00 on the four identical 8B GPUs; 3.6×10^{-3} on the mixed H100/H200 32B pods, 7x below the sham floor; every arm is compared against a baseline from its own machine.

C Deviations

D1: the published Qwen3-32B lens ships as a running-sum accumulator (`{jacobian_sum, n_done=80}`); the loader gained the shim $J = \text{sum}/n$. Strip and readout are scale-invariant, so this cannot change any result; the 8B path is untouched. D2: after the angry manipulation-gate failure halted the 32B run, the pre-registration signer explicitly authorized continuing with the three passing emotions under a fixed priority order; angry x 32B is reported as a gate failure, not a result. Both entries were written before any 32B outcome data existed. Post-run record: the three-emotion re-run passed every gate (same hit counts, same dose grid); the descriptive angry arena arms used the four-emotion run's vector packs (identical extraction, seed-fixed shams); all 22 planned arena cells and the 280-report battery completed.

D Instruments and compute

Arena categories with two examples each are listed in `instruments/arena64.json`. The 1-10 scale is read in two passes when "10" tokenizes as two tokens. Report probes ask for state, not emotion words. Compute: one RTX 4090 for the 8B pilot and confirmation shards (4 pods), two H100 and one H200 for 32B; story generation ~\$15 of API batch; total GPU spend ~\$110. Blind scoring: 664 sentences (340 8B confirmation, 44 dose sweep, 280 32B confirmation); one refusal total. The passive study (Section 5) adds one RTX 4090 for ~6 hours (~\$4) and 2,560 blind-scorer calls (1,920 turn labels, 640 forked reports, zero API errors); its harness, episode manifest, run outputs and analysis live in the repository's `gauge/` directory.